

Legacy Export Data-Health Audit

NorthLedger Insights - automated analysis

LEGACY EXPORT DATA-HEALTH AUDIT

1.

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1. SCOPE AND METHOD

Source: single table legacy_export. Size: 20,488 rows x 13 columns. All 13 columns are stored as text, so every value was profiled with pattern checks rather than type assumptions. Every count in this report comes from a query actually run against the table during this audit. The audit was read-only; no rows were modified. Coverage: 100% of rows were included in every census below (no sampling).

2. TABLE PROFILE

Columns: Collision ID, CRASH DATE, Crash Time, Boro, Zip (column name contains a trailing space), LATITUDE, LONGITUDE (column name contains a trailing space), On Street, number of persons injured, number of persons killed, CONTRIBUTING FACTOR, Vehicle Type 1, Vehicle Type 2.

Row count: 20,488. Unique Collision IDs after trimming and float-suffix removal: 20,000.

Content totals: 12,493 persons injured; 31 persons killed; 2,568 distinct street names; 14,911 distinct coordinate pairs; 356 distinct crash dates spanning March to June 2026.

3. FINDINGS WITH EXACT COUNTS

F1. DUPLICATES. 488 excess duplicate rows (20,488 rows minus 20,000 unique IDs), i.e. 2.4% of the table. All 488 duplicate ID groups are byte-identical across all 13 columns; zero groups contain conflicting values. Deduplication is therefore safe and mechanical.

F2. ID FORMAT CHAOS. 1,997 Collision IDs are padded with spaces (9.7%). 6,150 IDs carry a trailing ".0" from a float export (30.0%). Up to 8,147 cells (39.8%) need normalization. After trim and suffix removal the distinct-ID count stays at 20,000, confirming these are formatting artifacts, not distinct records.

F3. MIXED DATE FORMATS. Four formats cover 100% of the 20,488 dates:

MM/DD/YYYY	6,095 rows (29.8%)
DD-Mon-YY (15-Apr-26)	5,131 rows (25.1%)
ISO datetime	5,088 rows (24.8%)
ISO date	4,174 rows (20.4%)

16,314 rows (79.6%) are not in canonical ISO date form. All values matched a known pattern: zero unparseable dates, zero null-likes, zero empty values.

F4. MIXED TIME FORMATS. Four patterns cover 100% of the 20,488 times:

HH:MM 24h	7,203 rows (35.2%)
12-hour AM/PM (5:30 PM)	5,152 rows (25.1%)
compact HHMM (0733)	5,031 rows (24.6%)
single-digit-hour 24h (0:49)	3,102 rows (15.1%)

13,285 rows (64.8%) do not already match a zero-padded 24h HH:MM string. All values are parseable; zero null-likes.

F5. BORO. 4,977 rows are NULL (24.3%). 6,284 rows have leading or trailing spaces. 3,930 rows are lowercase: brooklyn 1,344; queens 979; manhattan 796; bronx 665; staten island 146. 20 raw spellings collapse to 5 real boroughs after trim plus uppercase: 15 phantom categories created by padding and casing alone.

F6. ZIP. 4,980 invalid values (24.3%): 1,688 empty (8.2%), 1,624 "N/A" (7.9%), 1,668 "0" (8.1%). A further 1,586 valid values are padded. 553 raw distinct values collapse to 380 after trim. The "0" values look like a system placeholder, not a real code.

F7. ON STREET. 6,595 rows empty (32.2%). 47 rows use non-canonical casing; 2,583 raw distinct values collapse to 2,568 (15 casing-only variants).

F8. COORDINATES. 626 rows (3.1%) are unusable: 167 empty LATITUDE/LONGITUDE pairs (0.8%) and 459 pairs set to 0.0000000 / 0.0000000 (2.2%). Missingness is perfectly symmetric: zero rows have one coordinate without the other. All remaining coordinates fall inside the NYC bounding box (latitude 40.4 to 41.0, longitude -74.3 to -73.5). 14,911 distinct coordinate pairs.

F9. NUMBER OF PERSONS INJURED. 2,085 values padded. 6,053 values are float-style text (e.g. "0.0", "1.0") or otherwise need numeric coercion. 35 raw distinct values collapse to 27 after trim. 5 rows report more than 10 injured (plausible multi-casualty events, but worth a spot check). Column total: 12,493.

F10. NUMBER OF PERSONS KILLED. 6,033 rows missing (29.4%): 3,035 empty (14.8%) plus 2,998 "N/A" (14.6%). 1,171 values padded. 8 raw distinct values collapse to 6. Missing values almost certainly mean zero fatalities and should be imputed to 0 after confirmation with the source system. Column total: 31.

F11. CONTRIBUTING FACTOR. 92 rows NULL (0.4%). 5,526 rows read "Unspecified" in some casing (27.0%) - a real value but zero information. 14,362 rows (70.1%) are not in canonical casing. 103 raw distinct labels collapse to 58 canonical ones: 45 labels exist purely as casing variants. Top canonical values: UNSPECIFIED 5,526; DRIVER INATTENTION/DISTRACTION 4,856; FAILURE TO YIELD RIGHT-OF-WAY 1,232; FOLLOWING TOO CLOSELY 1,093.

F12. VEHICLE TYPES. Vehicle Type 1: 350 NULL (1.7%) plus 1 "N/A"; 13,433 rows (65.6%) non-canonical casing; 209 raw values collapse to 119 (90 casing-only variants). Vehicle Type 2: 6,875 NULL (33.6%) plus 4 "N/A" (expected for single-vehicle crashes); 8,730 rows non-canonical casing; 229 raw values collapse to 129 (100 casing-only variants).

F13. SCHEMA HYGIENE. Two column names contain a trailing space ("Zip ", "LONGITUDE "). Naming mixes three conventions in one table (ALL-CAPS, Title Case, lowercase sentence). Every column, including identifiers and counts, is stored as text.

4. ISSUE MATRIX (row counts out of 20,488; "-" means not applicable)

Column	Missing/invalid	Needs reformat	Casing issues	Padding	Variant collapse
Collision ID	0	8,147	0	1,997	20,000 unique after norm
CRASH DATE	0	16,314	0	0	4 formats
Crash Time	0	13,285	0	0	4 patterns
Boro	4,977	-	3,930 lower	6,284	20 to 5
Zip	4,980	-	0	1,586	553 to 380
LATITUDE / LONGITUDE	626 pairs	-	0	0	14,911 pairs
On Street	6,595	-	47	0	2,583 to 2,568
persons injured	0	6,053	0	2,085	35 to 27
persons killed	6,033	0	0	1,171	8 to 6
CONTRIBUTING FACTOR	92 (+5,526 unspec)	0	14,362	0	103 to 58
Vehicle Type 1	351	0	13,433	0	209 to 119
Vehicle Type 2	6,879	0	8,730	0	229 to 129

Cell-level damage: approximately 116,000 problematic cells out of 266,344 total cells (about 43 to 44 percent).

5. CLEANING PLAN AND EFFORT

TIER 1 - MECHANICAL, SCRIPTABLE. Estimated 1.5 to 2.5 analyst-days to build plus 0.5 day to validate. Covers roughly 90% of flagged cells.

1. Deduplicate on normalized Collision ID: removes 488 rows, leaving 20,000. Safe: all duplicate groups byte-identical.
2. Normalize Collision ID: trim, strip trailing ".0". 8,147 cells.
3. Parse CRASH DATE with a four-pattern parser to ISO date. 16,314 cells.

4. Parse Crash Time with a four-pattern parser to zero-padded 24h HH:MM. 13,285 cells.
5. Trim all text columns. About 12,000 cells (Boro 6,284, Zip 1,586, injured 2,085, killed 1,171, ID 1,997).
6. Map categorical values to canonical codes via uppercase plus a lookup table (Boro, Contributing Factor, Vehicle Type 1 and 2). About 37,000 cells.
7. Coerce injured and killed to integers. 6,053 cells.
8. Rename columns to a single convention with no trailing spaces; retype ID, date, time, and count columns.

TIER 2 - POLICY DECISIONS, THEN SCRIPTED. Estimated 2 to 4 hours of decisions plus 0.5 day of rule writing.

1. Impute killed = 0 for the 6,033 missing values (confirm the zero-deaths assumption with the source system first).
2. Standardize "Unspecified" factor: keep as an explicit UNKNOWN category (5,526 rows) rather than null.
3. Convert Zip "0" (1,668) to NULL alongside empty and "N/A", creating a documented 4,980-row null population.
4. Fix the canonical casing policy (recommend uppercase codes for Boro and vehicle types, title case for street names).
5. Spot-check the 5 rows reporting more than 10 injured.

TIER 3 - NOT RECOVERABLE FROM THE DATA ALONE. Flag, do not fabricate.

1. 4,977 rows with no Boro, 6,595 rows with no street, 626 rows with dead coordinates. A portion may be back-fillable from the surviving coordinates via a spatial join against a borough and ZIP boundary file (not quantified in this audit).
2. 5,526 "Unspecified" factors are information that was never captured.

EFFORT SUMMARY

Tier 1 (mechanical transforms)	1.5 to 2.5 analyst-days
Tier 2 (policy plus rules)	0.5 to 1 analyst-day
Tier 3 (optional spatial backfill)	1 to 2 analyst-days
TOTAL to a validated clean table	3 to 5 analyst-days

EXPECTED END STATE after Tiers 1 and 2: 20,000 rows, zero duplicates, 100% ISO dates, uniform 24h times, zero padding, one canonical casing per category, integer injury counts, and a fully populated killed column. Residual nulls that cannot be manufactured away: Boro 4,977 (24.3%), Zip 4,980 (24.3%), On Street 6,595 (32.2%), coordinates 626 (3.1%), Factor 92 NULL plus 5,526 unspecified, Vehicle Type 1 351, Vehicle Type 2 6,879.

6. LIMITATIONS

- Read-only audit of one table. No lineage, no cross-check against the system of record, and no foreign-key targets were available.
- Every column is stored as text, so only pattern-level errors could be detected. A semantically wrong but well-formed date or coordinate would pass these checks.
- Slash dates were read as MM/DD/YYYY. All observed slash dates are consistent with this reading (first component never exceeds 12), but the data alone cannot fully exclude a DD/MM interpretation.
- Zip "0" values and 0.0/0.0 coordinates were treated as placeholder defaults. If they are legitimate encodings, the invalid counts change accordingly.
- Casing variants were assumed to represent the same entity (for example "sedan" equals "SEDAN"). The 90 to 100 variant collapses in vehicle types rest on that assumption and should be eyeballed once the canonical lookup table is built.
- Missing "number of persons killed" was assumed to mean zero. If it means "unknown", the 31-fatality total is understated and the imputation policy in Tier 2 must change.
- Duplicate detection used Collision ID as the key. Two genuinely different crashes that share a corrupted ID would not be flagged.
- The 5 rows with more than 10 injured were counted, not manually verified.

- Effort estimates assume scripted transforms with a review pass. Manual cleaning would be substantially more expensive.
- The potential for coordinate-based backfill of missing Boro and Zip values was noted but not quantified in this audit.